MIL-C-63277

9 March 1978
SUPERSEDING
WVTPD 1051
7 July 1971

MILITARY SPECIFICATION

COOLANT, GUIDED BORING: SULFURIZED FATTY AND MINERAL OIL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of sulfurized fatty-mineral oil with required additives for use as coolant and lubricant in the guided boring of cannon tubes.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-46148(MR) - Fatty Oil (for Metal-Working Lubricants)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-LCB-SAS, Watervliet Arsenal, Watervliet, NY 12189 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 9150

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STANDARDS

Federal

Fed. Standard 313 - Material Safety Data Sheets, Preparation and Submission of
Fed. Test Method Std. No. 791 - Lubricants, Liquid Fuels, and Related Products; Methods of Testing

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-290 - Packaging, Packing, and Marking of Petroleum and Related Products

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) TEST METHODS

ASTM D 92 - Flash and Fire Points by Cleveland Open Cup
ASTM D 95 - Water in Petroleum and Other Bituminous Materials
ASTM D 97 - Pour Point
ASTM D 270 - Sampling Petroleum and Petroleum Products
ASTM D 445 - Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities)
ASTM D 1662 - Active Sulfur in Cutting Fluids

(The ASTM test methods listed above are included in Parts 23 and 24 of the ASTM Book of Standards and are available individually for sale.)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS

3.1 Material. The cutting fluid shall be a blend of fatty oil and refined mineral oil in whatever proportions and with whatever additives that are necessary to meet the requirements of Table I.

3.2 Physical and chemical requirements.

3.2.1 Fatty oil blending stock. The fatty oil blending stock shall meet the requirements of MIL-F-46148(MR).

3.2.2 Mineral oil blending stock. Physical and chemical properties of the mineral oil blending stock shall be such as to permit the finished cutting fluid to meet the requirements of Table I.

3.2.3 Finished cutting fluid. The finished cutting fluid shall meet the requirements of Table I.

TABLE I. Physical and chemical requirements for finished cutting fluids

Properties	
Fatty oils: percent, min.	7.0
Active sulfur: percent, min.	0.6
Viscosity, Kinematic, at 24.0°C (75°F): centistokes	40-43
Viscosity, Kinematic, at 40°C (140°F): centistokes, min.	19
Pour point: °C max. (°F max.)	-1 (30)
Flash point: °C min. (°F min.)	160 (320)
Sediment & phase separation	None allowed (Observed visually without magnification)
Water content: percent by volume, max.	0.1

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3.3 Toxicity. The material shall have no adverse effects on the health of personnel when used for its intended purposes. Questions pertaining to toxicity shall be referred by the procuring activity to the appropriate departmental medical service (see 6.3) which will act as advisor to the procuring agency (see 4.6.2).

3.3.1 The coolant shall not contain nitrites or amines (see 4.6.2.2).

3.4 Workmanship. The cutting fluids shall be homogeneous, free-flowing, and free from objectionable odors.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot.

4.2.1 Bulk lot. An indefinite quantity of a homogeneous blend of one grade of cutting fluid offered for acceptance in a single, isolated container; or manufactured in a single plant run (not exceeding 24 hours) through the same processing equipment with no change in the ingredient materials.

4.2.2 Packaged lot. An indefinite number of drums or other unit containers of identical size and type, offered for acceptance at one time, and filled with a homogeneous blend of one grade of cutting fluid from a single, isolated container; or filled with a homogeneous mixture of one type of concentrate manufactured in a single plant run (not exceeding 24 hours), through the same processing equipment, with no change in the ingredient materials.

4.3 Sampling.

4.3.1 Sampling for examination of filled containers. Take a random sample of filled containers from each lot in accordance with MIL-STD-105, at inspection level II and acceptable quality level (AQL) = 2.5 percent defective.

4.3.2 Sampling for tests. Take samples for tests in accordance with ASTM method D 270.

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4.4 Inspection. Perform inspection in accordance with method 9601 of Fed. Test Method Std. No. 791.

4.4.1 Examination of filled containers. Examine samples taken as specified in 4.3.1 for conformance with MIL-STD-290 with regard to fill, closure, sealing, leakage, packaging, packing, and marking requirements. Reject any container having one or more defects or is underfilled. If the number of defective or underfilled containers exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, reject the lot represented by the sample.

4.5 Classification of tests. All tests are quality conformance tests.

4.6 Test methods. Perform tests and evaluations in accordance with table II.

TABLE II. Test methods

	Test method No. Fed. Std. 791	Test Method No. ASTM
Fatty oil content	3110	
Active sulfur		D 1662
Viscosity, kinematic		D 445
Pour point		D 97
Flash point		D 92
Sedimentation & Phase separation (see 4.6.1)		
Water content		D 95
Toxicity (see 4.6.2)		

4.6.1 Sedimentation and phase separation. Pour a sample of the oil as received into a clean, dry, stoppered 30-ml (4-oz) bottle and allow it to stand for 48 hours at 25° ± 3°C (77° ± 5°F). At the end of the 48 hour period, examine the sample visually for sedimentation or phase separation with the aid of a 40 watt incandescent bulb held approximately 150 millimeters behind and below the sample bottle.

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4.6.2 Toxicity. The supplier shall furnish the procuring activity with the toxicological data and formulations necessary to evaluate the physiological effects of the product. The data shall include evidence that the product is not injurious to human health when it is used as intended (see 6.1). Such evidence may be in the form of a toxicological study or in some other form acceptable to the procuring activity. Formulations shall include the percentages and standard chemical names of all constituent materials. The procuring activity shall have final authority for determining the acceptability of information or other evidence furnished by a potential supplier relative to the safety of his product.

4.6.2.1 Toxicological data shall include appropriate Material Safety Data Sheets (US Dept. of Labor Form OSHA-20 or equivalent) described in Federal Standard 313A Material Safety Data Sheets, Preparation and Submission of.

4.6.2.2 Certification that the coolant does not contain nitrites or amines shall be submitted by the manufacturer. Certification shall consist of a notorized statement that the fluid offered for acceptance does not contain nitrites or amines. The certification shall contain the name and address of the manufacturer, the trade name of the coolant, the name and address of the laboratory where the tests were made, and the date of these tests. The certification shall be signed by a responsible official of the manufacturer.

5. PACKAGING

5.1 Packaging, packing and marking. Unless otherwise specified in the contract (see 6.2), packaging, packing, and marking shall be in accordance with MIL-STD-290.

6. NOTES

6.1 Intended use.

6.1.1 This sulfurized fatty oil is intended for use in the guided boring of large caliber gun tubes (75mm thru 8 inch).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of cutting fluid required.

c. Type and size of container (see 5.1).

d. Level of packaging and level of packing required (see 5.1).

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6.3 Departmental medical services. Questions concerning the toxicity of commercial cutting fluids offered for sale to the Government should be referred to the appropriate departmental medical service in accordance with 3.3 and 4.6.2. Departmental medical services are as follows:

Army: Office of Surgeon General
Department of the Army
ATTN: HQDA (DASG-HCL)
Washington, D.C. 20310

Navy: Chief, Bureau of Medicine and Surgery
Bldg. 4, Potomac Annex
Code 4631
Department of the Navy
Washington, D.C. 20372

Air Force: Office of the Surgeon General
Directorate of Health Care Support
H.Q. USAF (AF/SGSL)
Forrestal Building
Washington, D.C. 20314

6.4 Supersession data. This specification includes the requirements of Purchase Description WVTPD 1051 dated 7 July 1971.

Custodians:
Army-MR
Navy-OS
Air Force-68

Review activities:
Army-AR
Navy-AS-SA
Air Force-68

User activities:
Army-AR
Navy-SH

Preparing activity:
Army-AR

Project No. 9150-0501

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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A. GIVE PARAGRAPH NUMBER AND WORDING

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